Infinitesimal Calculus 2, Simulation exam

2025, Reichman university.

Do not write at the back of the papers!!

Lecturer: Dr. Yossi Shamai. Duration: 3 hours.

Instructions:

The following exam has 5 questions. You must answer exactly 4 questions. Each question is worth 25 credits. There is no choice between different items of the same question.

Unexplained answers will not receive any credit. You must state all of the conditions for every theorem and/or claim. You do not have a choice between different items of the same question. It is forbidden to rely on any material that was not learned in the course. You must write your solutions only in the designated areas. Do not write at the back of the papers!

1. **(25 points)**

1.1. (17 points) Let $(a_n)_{n=1}^{\infty}$ be a sequence of real numbers.

1.1.1. (2 points) Define the following term: the series $\sum_{n=1}^{\infty}(-1)^{n+1}a_n$ is a Leibniz series.

Solution: We say that the series $\sum_{n=1}^{\infty}(-1)^{n+1}a_n$ is a Leibniz series if the following conditions hold:

(i) $a_{n+1} \leq a_n$ for every $n \in \mathbb{N}$.

(ii) $\lim_{n\to\infty} a_n = 0$.

1.1.2. (15 points) Suppose that the series $\sum_{n=1}^{\infty}(-1)^{n+1}a_n$ is a Leibniz series. Prove that the series converges to a value $s \in \mathbb{R}$, and that $a_1 - a_2 \leq s \leq a_1$.

Solution:

See lectures.

1.2. (8 points) Compute the value of the following series, or prove that it doesn't converge in the extended sense:

$$\sum_{n=1}^{\infty}\left(\frac{1}{2n-1}+\frac{1}{2n}-\frac{1}{n}\right)$$

Solution:

We know that

$$1-\frac{1}{2}+\frac{1}{3}-\frac{1}{4}+\cdots=\ln(2)$$

It follows that

$$\sum_{n=1}^{\infty}\left(\frac{1}{2n-1}+\frac{1}{2n}-\frac{1}{n}\right)=\sum_{n=1}^{\infty}\left(\frac{1}{2n-1}-\frac{1}{2n}\right)=\left(1-\frac{1}{2}\right)+\left(\frac{1}{3}-\frac{1}{4}\right)+\dots$$

$$\overset{\text{Parentheses test}}{=} 1-\frac{1}{2}+\frac{1}{3}-\frac{1}{4}+\dots=\ln(2)$$

2. **(25 points)**

2.1. (15 points) Prove that for every $-1 \le x \le 1$ we have

$$\arctan(x)=\sum_{n=0}^{\infty}\frac{(-1)^n x^{2n+1}}{2n+1}$$

Solution: See lectures.

2.2. (10 points) Compute the following limit, or prove that it doesn't exist in the extended sense

$$\lim_{n\to\infty}\left[n^{0.6}\left(n^2+1\right)^{0.7}-n^2\right]$$

Hint: Consider the function defined by $f(x)=x^{0.7}$ for every $x>0$.

Solution:

Consider the function defined by $f(x)=x^{0.7}$ for every $x>0$. By A.O.D., f is differentiable and $f'(x)=0.7\cdot x^{-0.3}$ for every $x>0$. We now compute

$$\begin{aligned}\lim_{n\to\infty}\left[n^{0.6}\left(n^2+1\right)^{0.7}-n^2\right]&=\lim_{n\to\infty}n^2\left[\frac{\left(n^2+1\right)^{0.7}}{n^{1.4}}-1\right]=\lim_{n\to\infty}n^2\left[\left(\frac{n^2+1}{n^2}\right)^{0.7}-1\right]\\&=\lim_{n\to\infty}n^2\left[\left(1+\frac{1}{n^2}\right)^{0.7}-1\right]=\lim_{n\to\infty}\frac{\left(1+\frac{1}{n^2}\right)^{0.7}-1}{\frac{1}{n^2}}\\&\overset{\text{Heine}}{=}\lim_{h\to0}\frac{(1+h)^{0.7}-1}{h}=f'(1)=0.7\end{aligned}$$

3. **(25 points)**

3.1. (13 points) Consider the sequence $(a_n)_{n=1}^{\infty}$ defined by the recursive formula

$$\begin{cases}a_{n+1}=a_n+\frac{1}{a_n} & \forall n\in\mathbb{N}\\ a_1=1\end{cases}$$

Prove that $\lim\limits_{n\to\infty}a_n=\infty$.

Solution:

We first prove that $\forall n\in\mathbb{N}$ we have $a_n>0$, and therefore the sequence $(a_n)_{n=1}^{\infty}$ is well-defined.

We prove this claim by induction:

Base case: $a_1>0$.

Induction step: Let $n\in\mathbb{N}$ such that $a_n>0$. Then $a_{n+1}=a_n+\frac{1}{a_n}>0$.

We now prove that the sequence $(a_n)_{n=1}^{\infty}$ is increasing. Let $n \in \mathbb{N}$. Then

$$a_{n+1} = a_n + \frac{1}{a_n} > a_n$$

ATC that $\lim\limits_{n\to\infty} a_n \neq \infty$. As $(a_n)_{n=1}^{\infty}$ is increasing, then there exists an $L \in \mathbb{R}$ such that

$$L = \lim_{n\to\infty} a_n \overset{\text{increasing}}{=} \sup\{a_n \,:\, n \in \mathbb{N}\}.$$

Note that $L \overset{\text{upper bound}}{\geq} a_1 = 1$ and therefore $L \neq 0$.

It follows that

$$L = \lim_{n\to\infty} a_{n+1} = \lim_{n\to\infty}\left[a_n + \frac{1}{a_n}\right] \overset{\text{AOL}}{=} L + \frac{1}{L}$$

Thus $\frac{1}{L} = 0$. Contradiction.

3.2. (12 points) Let $f : (0,2) \to \mathbb{R}$ be a differentiable function. Suppose that the limit $\lim\limits_{x\to 2^-} f(x)$ does not exist in the extended sense. Prove that there exists a $c \in (0,2)$ for which $f'(c) = 0$.

Solution:

ATC that $f'(x) \neq 0$ for every $0 < x < 2$.

Suppose first that $f'(x) > 0$ for every $x \in (0,2)$. Then f is strictly increasing, and therefore $\lim\limits_{x\to 2^-} f(x)$ exists in the extended sense and

$$\lim_{x\to 2^-} f(x) = \sup\{f(x) \,:\, 0 < x < 2\}$$

Contradiction.

Suppose now that $f'(x) < 0$ for every $x \in (0,2)$. Then f is strictly decreasing, and therefore $\lim\limits_{x\to 2^-} f(x)$ exists in the extended sense and

$$\lim_{x\to 2^-} f(x) = \inf\{f(x) \,:\, 0 < x < 2\}$$

Contradiction.

We conclude that there exists two points $0 < x_1 < x_2 < 2$ for which $f'(x_1) \cdot f'(x_2) < 0$. As f is differentiable on $[x_1, x_2]$, then by Darboux's theorem there exists an $x_1 < x < x_2$ for which $f'(x) = 0$. Contradiction.

4. **(25 points)**

4.1. (9 points) Let $a, b \in \mathbb{R}$ such that $a < b$ and let f be a function that is defined on $[a,b]$. Suppose that f^2 is integrable on $[a,b]$ and that $\int_a^b f^2(x)\,dx = 0$. Let $x_0 \in (a,b)$. Prove that if f is continuous at x_0, then $f(x_0) = 0$.

Solution:

Consider the function defined by $F(x) = \int_a^x f^2(t)\,dt$, for every $a \leq x \leq b$. Note that $0 \leq F(x) \leq \int_a^b f^2(x)\,dx$ for every $a \leq x \leq b$, implying that $F(x) = 0$ for every $x \in [a,b]$. As f is continuous at x_0, then f^2 is continuous at x_0. Thus, by the Fundamental theorem of calculus, $f^2(x_0) = F'(x_0) = 0$, which implies that $f(x_0) = 0$.

4.2. (9 points) Consider the function defined by $f(x) = \int_{3x}^{0} \frac{\arctan(t)}{2+t^4}\,dt$ for every $x > 0$. Compute $f'\left(\frac{1}{3}\right)$.

Solution:

First, as the function $\frac{\arctan(t)}{2+t^4}$ is continuous in $\mathbb{R}$ then it is integrable on every closed interval, and therefore f is well defined.

Note that for every $x \in \mathbb{R}$ we have

$$f(x) = \int_{3x}^{0} \frac{\arctan(t)}{2+t^4}\, dt = -\int_{0}^{3x} \frac{\arctan(t)}{2+t^4}\, dt$$

As the function $\frac{\arctan(t)}{2+t^4}$ is continuous in $\mathbb{R}$ then we know that for every $x \in \mathbb{R}$ we have

$$f'(x) = -3 \cdot \frac{\arctan(3x)}{2+(3x)^4}$$

Plugging $x = \frac{1}{3}$ into the above equation gives $f'\left(\frac{1}{3}\right) = -3\frac{\arctan(1)}{3} = -\frac{\pi}{4}$.

4.3. (7 points) Prove or disprove the following statement: We have

$$1 - \frac{\pi^2}{3!} + \frac{\pi^4}{5!} - \frac{\pi^6}{7!} + \cdots = 0.$$

Solution:

We have for every $x \in \mathbb{R}$, $\sin(x) = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \ldots$. Thus

$$0 = \sin(\pi) = \pi - \frac{\pi^3}{3!} + \frac{\pi^5}{5!} - \cdots = \pi\left[1 - \frac{\pi^2}{3!} + \frac{\pi^4}{5!} - \frac{\pi^6}{7!} + \ldots\right]$$

Dividing both sides by π gives the result.

5. **(25 points)**

5.1. (13 points) Let $(a_n)_{n=1}^{\infty}$ be a sequence of positive numbers and let $p \in \mathbb{R}$. Consider the sequence defined by

$$b_n = \frac{\ln\left(\frac{1}{a_n}\right)}{\ln(n)}, \qquad \forall n \in \mathbb{N}$$

Suppose that $\lim_{n\to\infty} n^{p-b_n} = 1$. Prove that the series $\sum_{n=1}^{\infty} a_n$ is convergent if and only if $p > 1$.

Solution:

Note that as

$$b_n = \frac{\ln\left(\frac{1}{a_n}\right)}{\ln(n)}, \qquad \forall n \in \mathbb{N}$$

then $b_n \ln(n) = \ln\left(\frac{1}{a_n}\right) = -\ln(a_n)$ and therefore

$$-b_n \ln(n) = \ln(a_n) \Rightarrow \ln\left(n^{-b_n}\right) = \ln(a_n) \Rightarrow a_n = \frac{1}{n^{b_n}}$$

We now have

$$\lim_{n\to\infty} \frac{a_n}{\left[\frac{1}{n^p}\right]} = \lim_{n\to\infty} \frac{\left[\frac{1}{n^{b_n}}\right]}{\left[\frac{1}{n^p}\right]} = \lim_{n\to\infty} n^{p-b_n} = 1$$

Thus, the series converges together with the p-harmonic series, so it converges if and only if $p > 1$.

5.2. (12 points) Compute the value of the following limit, or prove that it doesn't exist in the extended sense:

$$\lim_{x\to\infty} x^2\left[2\ln(x) - \ln\left(x^2+1\right)\right]$$

Solution:

We have

$$\lim_{x\to\infty} x^2\left[2\ln(x)-\ln\left(x^2+1\right)\right] = \lim_{x\to\infty}\frac{\ln\left(x^2\right)-\ln\left(x^2+1\right)}{\left[\frac{1}{x^2}\right]} = \lim_{x\to\infty}\frac{\ln\left(\frac{x^2}{1+x^2}\right)}{\left[\frac{1}{x^2}\right]} = \lim_{x\to\infty}\frac{\ln\left(1-\frac{1}{1+x^2}\right)}{\left[\frac{1}{x^2+1}\right]\cdot\left[1+\frac{1}{x^2}\right]} \overset{(\star)}{=} -1$$

$$\lim_{x\to\infty}\frac{\ln\left(1-\frac{1}{1+x^2}\right)}{\left[\frac{1}{x^2+1}\right]} \overset{t=\frac{1}{1+x^2}}{=} \lim_{t\to 0}\frac{\ln(1-t)}{t} \overset{\frac{0}{0}}{=} \lim_{t\to 0}\frac{\left[-\frac{1}{1-t}\right]}{1} = -1 \qquad (\star)$$

<u>Alternative solution:</u> We have

$$\begin{aligned}\lim_{x\to\infty} x^2\left[2\ln(x)-\ln\left(x^2+1\right)\right] &= \lim_{x\to\infty}\frac{\ln\left(x^2\right)-\ln\left(x^2+1\right)}{\left[\frac{1}{x^2}\right]} \overset{\frac{0}{0}}{=} \lim_{x\to\infty}\frac{\frac{2x}{x^2}-\frac{2x}{x^2+1}}{\left[-\frac{2}{x^3}\right]}\\ &= \lim_{x\to\infty}\left[-x^2+\frac{x^4}{x^2+1}\right] = \lim_{x\to\infty}\frac{x^4-x^2\left(x^2+1\right)}{x^2+1}\\ &= \lim_{x\to\infty}\frac{-x^2}{x^2+1} = \lim_{x\to\infty}\frac{-1}{1+\frac{1}{x^2}} \overset{\text{AOL}}{=} -1\end{aligned}$$